

Day 9 - Prime Number Check ☐

Question

Ek number **N** diya gaya hai.

Check karo ki wo **Prime Number** hai ya nahi.

Prime Number

Prime number wo hota hai jo:

Sirf 2 numbers se divide hota hai

1 aur khud se

Example

2 → Prime

3 → Prime

5 → Prime

7 → Prime

4 → Not Prime

6 → Not Prime

8 → Not Prime

9 → Not Prime

Approach 1 - Brute Force

Logic

1 se N tak divide karke count karo kitne divisors hain.

Dry Run

N = 5

1 divides 5 ✓
2 divides 5 ✗
3 divides 5 ✗
4 divides 5 ✗
5 divides 5 ✓

Count = 2

Prime

Java Code

```
public class Day9 {  
    public static void main(String[] args) {  
        int n = 5;  
        int count = 0;  
        for(int i = 1; i <= n; i++) {  
            if(n % i == 0)  
                count++;  
        }  
        if(count == 2)  
            System.out.println("Prime");  
        else  
            System.out.println("Not Prime");  
    }  
}
```

Time Complexity

$O(N)$

Space Complexity

$O(1)$

Approach 2 - Better Approach

Logic

2 se N-1 tak check karo.

Agar koi number divide kar raha hai to prime nahi hai.

Java Code

```
public class Day9 {  
    public static void main(String[] args) {  
        int n = 17;  
        boolean prime = true;  
  
        if(n <= 1)  
            prime = false;  
  
        for(int i = 2; i < n; i++) {  
            if(n % i == 0) {  
                prime = false;  
                break;  
            }  
        }  
  
        if(prime)  
            System.out.println("Prime");  
        else  
            System.out.println("Not Prime");  
    }  
}
```

Time Complexity

$O(N)$

Space Complexity

$O(1)$

Approach 3 - Optimal Approach ☐

Important Interview Concept

Agar kisi number ka factor hai to uska pair factor bhi hota hai.

Example:

36

1 × 36

2 × 18

3 × 12

4 × 9

6 × 6

Factors $\sqrt{N}$ ke baad repeat hone lagte hain.

Isliye sirf:

2 se $\sqrt{N}$

tak check karna hota hai.

Dry Run

N = 49

$\sqrt{49} = 7$

Check:

2

3

4

5

6

7

$49 \% 7 = 0$

Not Prime

Java Code

```
public class Day9 {  
    public static void main(String[] args) {  
        int n = 29;  
        if(n <= 1) {  
            System.out.println("Not Prime");  
            return;  
        }  
        boolean prime = true;  
        for(int i = 2; i * i <= n; i++) {  
            if(n % i == 0) {  
                prime = false;  
                break;  
            }  
        }  
        if(prime)  
            System.out.println("Prime");  
        else  
            System.out.println("Not Prime");  
    }  
}
```

Time Complexity

$O(\sqrt{N})$

Space Complexity

$O(1)$

Interview Follow-Up Questions

Q1. Is 1 Prime?

No

Prime number ke exactly 2 factors hone chahiye.

1 ke sirf 1 factor hai

Q2. Smallest Prime Number?

2

Q3. Only Even Prime Number?

2

Baaki saare even numbers:

2 se divide ho jate hain

Isliye prime nahi hote.

Q4. Why $\sqrt{N}$ Optimization Works?

Example:

100

Factors:

1 × 100

2 × 50

4 × 25

5 × 20

10 × 10

Ek factor $\sqrt{N}$ ke baad aaya to doosra factor $\sqrt{N}$ se pehle already mil chuka hoga.

Isliye:

Sirf $\sqrt{N}$ tak check karna kaafi hai.

Interview Concept of the Day

Perfect Square Optimization

Instead of:

```
Math.sqrt(n)
```

Interviews me likho:

```
i * i <= n
```

Kyunki:

```
Math.sqrt()
```

har iteration me call hota hai.

Aur floating point calculation bhi karta hai.

Summary

Approach	Time	Space
Count Factors	$O(N)$	$O(1)$
Check till $N-1$	$O(N)$	$O(1)$
Check till $\sqrt{N}$ □	$O(\sqrt{N})$	$O(1)$

Most Asked Interview Test Cases

Test Case 1

`n = 2`

Output:

`Prime`

Test Case 2

`n = 1`

Output:

`Not Prime`

Test Case 3

`n = 17`

Output:

`Prime`

Test Case 4

`n = 49`

Output:

`Not Prime`

Test Case 5

`n = 97`

Output:

`Prime`

Practice Questions

Check manually:

11
15
19
25
29
37
49
53

Prime ya Not Prime?